Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI-enabled EV charging station information and charge planning platform with energy hub optimization			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communications		Electrical & Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Software	Internet SW	Heavy Electrical Equipment	Power conversion equipment
Concept of Proposed Technology	**Core concept** TNO proposes a modular software stack that increases the reliability and usefulness of live EV charging station information by combining protocol-level telemetry (OCPP), roaming data exchange (OCPI), predictive models and multi-objective optimization for routing and scheduling. [R1] [R2] [R3] [R4] [R10] [R11]			

Technical architecture (functional blocks)

1) Data ingestion and interoperability: OCPP and OCPI connectors mapped into a canonical EVSE data model, with data-quality checks and anomaly detection (stale status, inconsistent connector metadata, missing transactions) producing a per-EVSE confidence score. [R10] [R11]

2) Prediction: (i) availability and wait-time prediction per connector using probabilistic time-series and survival models; (ii) charging duration and power profile prediction using a battery Digital Twin based method with fast, low-training-cost prediction. [R3]

3) Optimization: (i) charge planning that trades off travel time, predicted wait and charging cost under uncertainty; (ii) fleet and depot charging schedules for heterogeneous vehicles and chargers with grid constraints and degradation-aware cost terms (calendar and cyclic ageing). [R1] [R2] [R4]

4) Energy hubs: hub-level energy management to size and operate chargers combined with local flexibility (stationary storage, on-site renewables) to mitigate grid congestion and reduce peaks. [R4] [R6] [R7]

5) Deployment: containerized microservices with documented APIs plus monitoring (service KPIs and model drift) for integration into the Korean partner's mobile and backend stack.

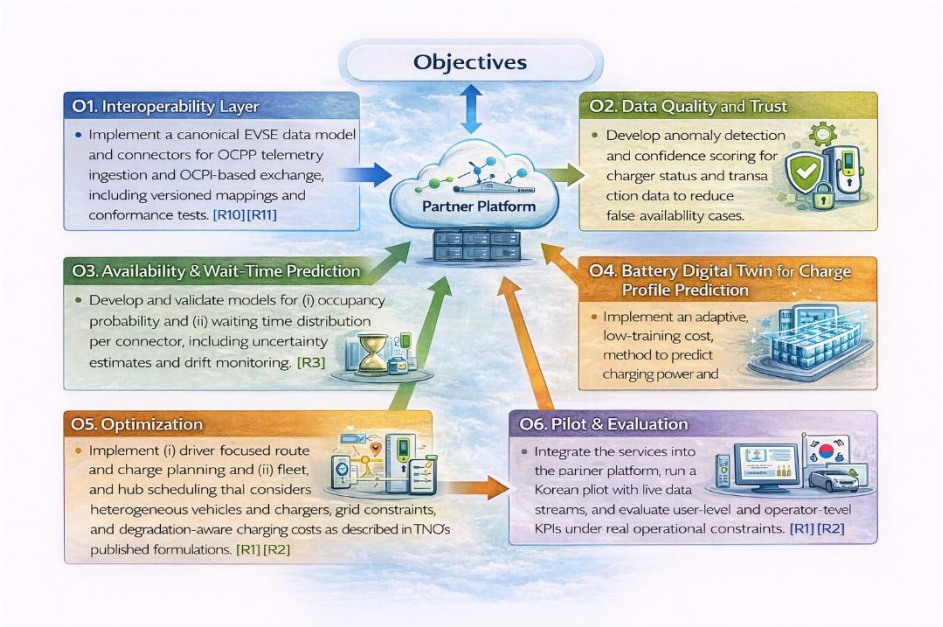
Why TNO

TNO Powertrains has published and validated methods for fleet charging optimization that address grid constraints and battery degradation, and has published battery Digital Twin based charge profile prediction and charge planning tool concepts for heavy-duty fleets. [R1] [R2] [R3] [R4]

TNO's broader competencies include battery monitoring and control and energy management algorithms in mobility contexts. [R8] [R9] [R13]

Rationale for Joint R&D Project	Why a joint R&D project is needed The Korean SME partner operates an EV charging station information and navigation service. Their performance depends on (i) correctness of live availability and (ii) decision quality for recommended charging stops under uncertainty. A joint R&D project with TNO is proposed because the following capabilities require specialized powertrain, battery and energy-systems expertise plus validated algorithms: 1) Degradation-aware charging and scheduling: optimization under grid and operational constraints with explicit ageing costs. [R1] [R2] [R4] 2) Battery Digital Twin based charging-duration prediction: adaptive charge profile prediction for improved charging-time and grid-load forecasting. [R3] 3) Energy hub and congestion management: methods and software concepts for charging energy hubs and megawatt charging hubs. [R6] [R7] 4) Standards-based interoperability: robust implementation of OCPP and OCPI semantics and version differences with conformance testing. [R10] [R11] Together, these elements enable a shift from static map information toward a predictive, optimization-driven platform that can reduce waiting and failed charge attempts while improving utilization and peak management.
Key Words	1. Charge planning 2. Charger availability prediction 3. Battery Digital Twin 4. Charging energy hubs 5. Degradation-aware optimization

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	Objectives  O1. Interoperability layer: implement a canonical EVSE data model and connectors for OCPP telemetry ingestion and OCPI-based exchange, including versioned mappings and conformance tests. [R10] [R11] O2. Data quality and trust: develop anomaly detection and confidence scoring for charger status and transaction data to reduce false-availability cases and improve operational observability. O3. Availability and wait-time prediction: develop and validate models for (i) occupancy probability and (ii) waiting time distribution per connector, including uncertainty estimates and drift monitoring. O4. Battery Digital Twin for charge profile prediction: implement an adaptive, low-training-cost method to predict charging power and duration for a given charger class and initial battery state, based on TNO's published work (Rath and Wilkins). [R3] O5. Optimization: implement (i) driver-focused route and charge planning and (ii) fleet and hub scheduling that considers heterogeneous vehicles and chargers, grid constraints and degradation-aware charging costs as described in TNO's published formulations. [R1] [R2] [R4] O6. Pilot and evaluation: integrate the services into the partner platform, run a Korean pilot with live data streams, and evaluate user-level and operator-level KPIs under real operational constraints.			

R&D Target Specifications	Target specifications (project targets) Data and latency: near real-time status ingestion for OCPP-connected charge points with end-to-end update latency target <= 60 seconds for status changes (where partner backends deliver telemetry). [R10] Charge time prediction: mean absolute percentage error (MAPE) target <= 5% on representative AC and DC sessions, with the calibrated Digital Twin approach targeting <= 2% on well-instrumented sites as reported. [R3] Wait-time prediction: mean absolute error target <= 5 minutes for the 50th percentile waiting time at busy sites, with calibrated uncertainty bands (for example, 50th and 90th percentile). Availability classification: AUC target >= 0.85 for 15 minute availability prediction at high-traffic sites (to be validated per network). Fleet schedule computation: target <= 3 minutes for 200 vehicles and 100 chargers on a standard server instance, aligned with real-time execution focus in the published scheduling approach. [R2] Hub peak reduction: target demonstrate >= 10% reduction in peak grid import in at least one pilot hub when local flexibility is available. [R4] [R6] Interoperability: OCPP (2.0.1 or newer where supported) and OCPI (2.2.x) compatibility in the integration layer. [R10] [R11] Deployment: containerized microservices with documented REST APIs and monitoring, suitable for integration into the partner stack.
R&D Plan	R&D plan (36 months, example) Months 0-6: Requirements, data governance and baseline KPIs; deliver data model and API specification including OCPP and OCPI mapping notes and test plan. [R10] [R11] Months 4-14: Build ingestion connectors and canonical EVSE store; implement data validation, anomaly detection and confidence scoring; deliver production-ready data service with monitoring. Months 6-20: Develop availability and wait-time prediction models; implement battery Digital Twin based charge profile predictor and calibration workflow (Rath and Wilkins). [R3] Months 10-26: Implement optimization engines for driver routing and for fleet and hub scheduling with grid limits and degradation-aware costs, building on TNO's published work. [R1] [R2] [R4] Months 18-32: Integrate services into partner platform; run staged Korean pilot rollout (shadow

	mode, beta, scaled pilot); iterate using operational feedback and model-drift monitoring. Months 30-36: Final validation and transfer package (documentation, training, maintenance plan); scientific output and benchmark methodology subject to confidentiality constraints.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TNO (research institute) - Reference architecture and interface specification for data, prediction and optimization services. - Model development and validation: availability and wait-time prediction; battery Digital Twin based charge profile prediction. [R3] - Optimization development: routing, fleet scheduling and energy hub operation with grid and degradation constraints. [R1] [R2] [R4] - Standards and interoperability: implement and test OCPP and OCPI mappings and conformance checks. [R10] [R11] - Pilot support and transfer: offline benchmarking, online monitoring (service KPIs and model drift), documentation and training.
Expected R&R of Korean SME Partner	Korean SME partner - Provide access to relevant charging station datasets and operational constraints (subject to agreements). - Integrate TNO services into production platform (backend and mobile UX) and operate pilot deployments in Korea. - Lead product requirements, user journeys and success metrics; handle go-to-market, customer support and scaling. - Coordinate pilot sites and stakeholders (charging point operators, permissions, field feedback).

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Expected outcomes and impact 1) Higher-quality charging station information: reduced false-availability events and improved user trust via confidence scoring and anomaly detection. 2) Better charging decisions: predictive availability, wait-time and charging-duration estimates enable route choices that minimize total trip time and charging risk. 3) Lower operational cost and reduced battery wear: degradation-aware scheduling can outperform naive charging, consistent with TNO's published formulations. [R2] [R4] 4) Operator and grid benefits: energy hub and fleet scheduling modules can reduce peaks and support congestion management. [R6] [R7] 5) Technology transfer: delivery of containerized services, documentation and training enabling deployment and maintenance by the Korean SME partner. 6) Dissemination and IP: joint publications and optional co-development of algorithms or joint IP depending on program terms and confidentiality constraints.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents

	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization